

Qatar-2b

R-1499

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_200227 · created 2026-08-01T14:38:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458906.88696 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.065 +/- 0.027
Transit depth (Rp/Rs)^2	0.42 +/- 0.34 [%]
Orbital Inclination (inc)	85.46 +/- 1.7
Airmass coefficient 1 (a1)	1.207 +/- 0.014
Airmass coefficient 2 (a2)	-0.137 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	1.17 %
Best Comparison Star	None
Optimal Aperture	10.12
Optimal Annulus	25.31
Transit Duration (day)	0.0599 +/- 0.0094

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.065 $\pm$ 0.027 vs 0.16327 (deviation 2.81 σ)
Duration fit vs published	0.0599 d vs 0.0767 d (deviation 1.38 σ)
Mid-time O-C	-1.8 min
Depth SNR	1.9
Residual scatter	1.17 %

Light curve

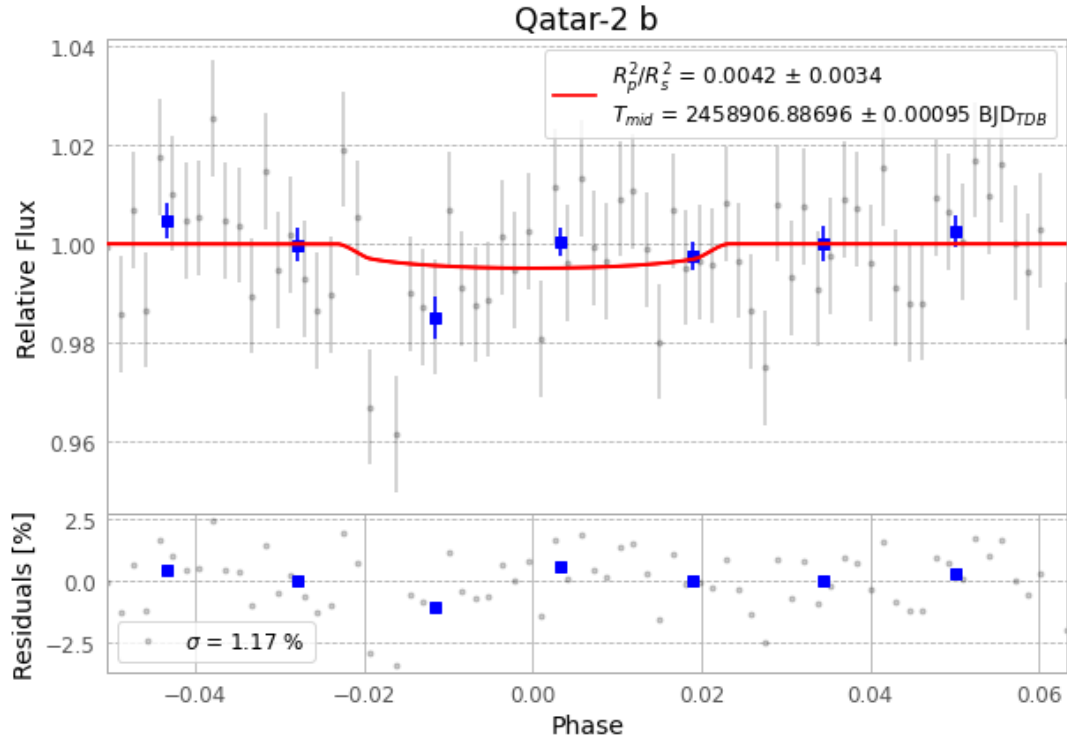


Figure 1 — FinalLightCurve_Qatar-2 b_2020-02-27.png (SHA-256 6e045680b497fc68...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [136, 410], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b5358bf469f764eb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), Qatar-2200227073312.FITS → Qatar-2200227111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-200227.log (858d8706d775...), FinalLightCurve_Qatar-2 b_2020-02-27.png (6e045680b497...), FinalLightCurve_Qatar-2 b_2020-02-27.csv (04e2d44a5d33...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 14:38Z.